Eine Woche, ein Beispiel

12.21. Hodge structure

Ref:

[Car17]: James Carlson, Stefan Müller-Stach, and Chris Peters. Period Mappings and Period Domains. 2nd ed. Cambridge University Press, 2017. https://doi.org/10.1017/9781316995846.
[Wells08]: Raymond O Wells. Differential Analysis on Complex Manifolds. Vol. 65. Graduate Texts in Mathematics. Springer New York, 2008. https://doi.org/10.1007/978-0-387-73892-5.
[JC97]: John Horton Conway. The Sensual (Quadratic) Form. Assisted by Francis Y. C. Fung. Vol. 26. Carus Math. Monogr. The Mathematical Association of America, 1997.
[Cox13]: David Cox A. Primes of the Form \(X^2+ny^2\). Fermat, Class Field Theory, and Complex Multiplication. 2nd ed. Pure Appl. Math., Wiley Ser. Texts Monogr. Tracts. John Wiley & Sons, 2013.

X: cpt Kähler mfld of dim n.

1. definition and basic example
2. integral bilinear forms

1. definition and basic example

Slogan: Hodge structure collects (nearly) all linear algebraic structures of X.

These linear algebraic structures are: $k = p+q$

① $H^k(X;\mathbb{Z})$

② $H^q(X;\Omega_X^p) \cong \{\text{harmonic } (p,q)\text{-forms}\}$

③ A comparison iso as $\mathbb{C}$-v.s. (J-action)

$$H^k(X;\mathbb{Z})\otimes_{\mathbb{Z}}\mathbb{C} \cong \bigoplus_{p+q=k} H^q(X;\Omega_X^p)$$

②′ A Hodge filtration

$$H^k(X;\mathbb{Z})\otimes_{\mathbb{Z}}\mathbb{C} \cong \underset{H^{0,k}}{H^k(\Omega_X^\cdot)} \supset \underset{H^{1,k-1}}{H^{k-1}(\Omega_X^{\geq 1})} \supset \underset{H^{2,k-2}}{H^{k-2}(\Omega_X^{\geq 2})} \supset \cdots \supset \underset{H^{k-1,1}}{H^1(\Omega_X^{\geq k})} \supset \underset{H^{k,0}}{0}$$

③′ Compatability with conj_J:

$$H^k(X,\mathbb{Z})\otimes_{\mathbb{Z}}\mathbb{C} = H^{k-p}(\Omega_X^{\geq p}) \oplus \overline{H^{p-1}(\Omega_X^{\geq k-p+1})}$$

Here, $\Omega_X^{\geq p} := \sigma_{\geq p}\Omega_X^\cdot$. We have

$$\begin{tikzcd}
 & & 0 \arrow[dr] & & 0 & & \\
 & & & \Omega_X^{\geq 1} \arrow[ur] \arrow[dr] & & & \\
0 \arrow[r] & \underline{\mathbb{C}} \arrow[r] \arrow[dr] & \Omega_X^0 \arrow[ur] \arrow[r, "d^1"] & \Omega_X^1 \arrow[r, "d^2"] \arrow[dr] & \Omega_X^2 \arrow[ur] \arrow[r, "d^3"] & \cdots & \\
 & & \Omega_X^\cdot = \Omega_X^{\geq 0} \arrow[ur] & & \Omega_X^{\geq 2} \arrow[ur] \arrow[dr] & & \\
 & 0 \arrow[ur] & & 0 \arrow[ur] & & 0 &
\end{tikzcd}$$

④ (polarization)

A bilinear form

$$\Lambda : H^k(X;\mathbb{C}) \otimes H^k(X;\mathbb{C}) \xrightarrow{\Lambda} H^{2k}(X;\mathbb{C}) \xrightarrow{\wedge \omega^{n-k}} H^{2n}(X;\mathbb{C}) \cong \mathbb{C}$$

$$\Lambda(\xi,\eta) := \int_X \xi \wedge \eta \wedge \omega^{n-k}$$

with two compatability conditions:

Assume ω is integral, i.e., X is proj

$$\begin{array}{ccccccc}
\Lambda : H^k(X;\mathbb{Z}) & \otimes & H^k(X;\mathbb{Z}) & \xrightarrow{\Lambda} & H^{2k}(X;\mathbb{Z}) & \xrightarrow{\wedge \omega^{n-k}} & H^{2n}(X;\mathbb{Z}) \cong \mathbb{Z} \\
 & \cap & & & \cap & & \cap \\
\Lambda : H^k(X;\mathbb{C}) & \otimes & H^k(X;\mathbb{C}) & \xrightarrow{\Lambda} & H^{2k}(X;\mathbb{C}) & \xrightarrow{\wedge \omega^{n-k}} & H^{2n}(X;\mathbb{C}) \cong \mathbb{C} \\
 & \cup & & & \cup & & \cong \\
\Lambda : H^q(X;\Omega_X^p) & \otimes & H^p(X;\Omega_X^q) & \xrightarrow{\wedge} & H^k(X;\Omega_X^k) & \xrightarrow{\wedge \omega^{n-k}} & H^n(X,\omega_X) \cong \mathbb{C}
\end{array}$$

with two positivity conditions:

$$\Lambda(\xi,\bar{\xi}) \in \mathbb{R} \quad \text{for} \quad \xi \in H^k(X;\mathbb{R}) \qquad \xi = \bar{\xi} \text{ here}$$

$$i^{p-q+k(k-1)}\Lambda(\xi,\bar{\xi}) \geq 0 \quad \text{for} \quad \xi \in H^k_{prim}(X;\mathbb{C}) \cap H^{p,q}$$

And get "=" $\Leftrightarrow$ $\xi = 0$.

⚠ These positivities are not contradiction to each other.

If $\xi \in H^k(X;\mathbb{R}) \cap (H^k_{prim}(X;\mathbb{C}) \cap H^{p,q})$, then $\xi \in \overline{H^{p,q}} = H^{q,p}$,

then $p = q$, i.e.

$\xi \in H^{2p}_{prim}(X;\mathbb{R}) \cap H^{p,p}$, $\quad i^{p-q+k(k-1)} = (-1)^p$.

Rmk. 1. Even though $\int_X \omega^n > 0$, one don't nec have

$$\Lambda(\xi, \bar{\xi}) = \int_X \xi \wedge \bar{\xi} \wedge \omega^{n-k} \geq 0 \qquad \text{for any } \xi \in H^k(X; \mathbb{R}).$$

For example, for a cplx K3 surface X,
$H^2(X;\mathbb{Z})$ has signature $(3,19)$, not $(0,22)$ or $(2,20)$

This is also a counterexample if one forgets "prim".

2. A de-Rham explanation for the following commutative diagram is here:
https://math.stackexchange.com/questions/29797/direct-proof-that-the-wedge-product-preserves-integral-cohomology-classes

$$\begin{tikzcd}
H^i(X;\mathbb{Z}) \otimes H^j(X;\mathbb{Z}) \arrow[r, "\cup"] \arrow[d] & H^{i+j}(X;\mathbb{Z}) \arrow[d] \\
H^i_{dR}(X;\mathbb{C}) \otimes H^j_{dR}(X;\mathbb{C}) \arrow[r, "\wedge"] & H^{i+j}_{dR}(X;\mathbb{C})
\end{tikzcd}$$

3. The positivity condition is in fact the generalized Hodge--Riemann bilinear relation,
which is derived from the hard Lefschetz theorem.
(But can be naively understood/remembered through the shape of differential forms)

For a ref, see:

https://arxiv.org/abs/1905.13636
[Car17, Thm 2.3.3]

[Wells08]: p161, Lem 1.2 $\longrightarrow$ p187, Thm 3.16 $\longrightarrow$ p207, Thm 6.3
provides coefficients $\qquad \int_X \xi \wedge *\bar{\xi} = \|\xi\|^2_{L^2} \geq 0$

4. The coefficient is somehow cumbersome,
because people use different normalizations.

① Some people use F as the integral Kähler form, where $\omega = \frac{i}{2\pi} F$;

Not clear about this myself, but when I check the ref, no people use this normalization.
So probably I'm just be too scared about the 2\pi i...

② Some people use $\qquad \xi, \xi' \in H^{p,q}, \ \eta \in H^{q,p}$

$$\begin{aligned}
b(\xi, \eta) &= (-1)^{\frac{1}{2}k(k-1)} \wedge (\xi, \eta) \\
h_c(\xi, \xi') = i^{p-q} b(\xi, \bar{\xi}') &= i^{p-q+k(k-1)} \wedge (\xi, \bar{\xi}') \\
Q(\xi, \eta) = (-1)^k b(\xi, \eta) &= (-1)^{\frac{1}{2}k(k+1)} \wedge (\xi, \eta)
\end{aligned}$$

in the literature.

Why do we use the coefficient $(-1)^{\frac{1}{2}k(k-1)}$?

Intuitive explanations:

$$
\begin{aligned}
\xi \wedge \bar{\xi} \wedge \omega^{n-k} &\approx (dz_1 \wedge \cdots) \wedge (d\bar{z}_{p+1} \wedge \cdots) \wedge (d\bar{z}_1 \wedge \cdots) \wedge (dz_{p+1} \wedge \cdots) \wedge \omega^{n-k} \\
&= (-1)^{\frac{1}{2}k(k-1)} (dz_1 \wedge d\bar{z}_1) \wedge \cdots \wedge (d\bar{z}_{p+1} \wedge dz_{p+1}) \wedge \cdots \wedge \omega^{n-k} \\
&= (-1)^{\frac{1}{2}k(k-1)} (-2i)^p (2i)^q (dx_1 \wedge dy_1) \wedge \cdots \wedge (dx_{p+1} \wedge dy_{p+1}) \wedge \cdots \wedge \omega^{n-k} \\
&= (-1)^{\frac{1}{2}k(k-1)} i^{q-p} 2^k (dx_1 \wedge dy_1) \wedge \cdots \wedge (dx_{p+1} \wedge dy_{p+1}) \wedge \cdots \wedge \omega^{n-k}
\end{aligned}
$$

Since
$$
\begin{aligned}
\Lambda(\xi, \eta) &= (-1)^k \Lambda(\eta, \xi), \\
b(\xi, \eta) &= (-1)^k b(\eta, \xi) \\
h_c(\xi, \xi') &= h_c(\bar{\xi}', \bar{\xi}) \\
Q(\xi, \eta) &= (-1)^k Q(\eta, \xi)
\end{aligned}
$$
we gets

Since
$$
\begin{aligned}
\overline{\Lambda(\xi, \eta)} &= \Lambda(\bar{\xi}, \bar{\eta}), \\
\overline{b(\xi, \eta)} &= b(\bar{\xi}, \bar{\eta}) \\
\overline{h_c(\xi, \xi')} &= h_c(\bar{\xi}, \bar{\xi}') \\
\overline{Q(\xi, \eta)} &= Q(\bar{\xi}, \bar{\eta})
\end{aligned}
$$
we gets

Therefore,
$$\overline{h_c(\xi, \xi')} = h_c(\xi', \xi)$$

$\rightsquigarrow$ h_c is a positive definite Hermitian form on $H^{p,q}$.

The Hodge module abstracts the all above properties.

Parallel, we have:

① A f.g. abelian gp $H_{\mathbb{Z}}$ $\qquad H_{\mathbb{C}} := H_{\mathbb{Z}} \otimes_{\mathbb{Z}} \mathbb{C}$

② Real v.s. $H^{p,q} \subset H_{\mathbb{C}}$ $\qquad$ for all $p+q=k$, $p,q \in \mathbb{Z}$

③ A dec (called Hodge dec)

$$H_{\mathbb{C}} = \bigoplus_{p+q=k} H^{p,q} \qquad \text{with } \overline{H^{p,q}} = H^{q,p}$$

②' A filtration (called Hodge filtration)

$$H_{\mathbb{C}} \supset \cdots \supset F^0 H_{\mathbb{C}} \underset{H^{0,k}}{\supset} F^1 H_{\mathbb{C}} \underset{H^{1,k-1}}{\supset} F^2 H_{\mathbb{C}} \supset \cdots \supset F^k H_{\mathbb{C}} \underset{H^{k,0}}{\supset} F^{k+1} H_{\mathbb{C}} \supset \cdots$$

③' Compatability with conjugation:

$$H_{\mathbb{C}} = F^p H_{\mathbb{C}} \oplus \overline{F^{k-p+1} H_{\mathbb{C}}}$$

④ (polarization)

A bilinear form $\quad k = p+q$

$$b : H^{p,q} \otimes H^{q,p} \longrightarrow \mathbb{C} \qquad b = (-1)^{\frac{1}{2}k(k-1)} \wedge$$

which extends to $b : H_{\mathbb{C}} \times H_{\mathbb{C}} \longrightarrow \mathbb{C}$ via zero extension.

$$b(H^{p,q} \times H^{q',p'}) \equiv 0 \quad \text{if} \quad (p,q) \neq (p',q')$$

i.e.,

$$b(F^p H_{\mathbb{C}}, F^{n-p+1} H_{\mathbb{C}}) \equiv 0 \qquad \text{(first bilinear relation)}$$

with positivity condition $\quad$ for all $\xi \in H^{p,q}$ nonzero

$$i^{p-q} b(\xi, \bar{\xi}) > 0 \qquad \text{(second bilinear relation)}$$

and integral condition

$$b(H_{\mathbb{Z}}, H_{\mathbb{Z}}) \subseteq \mathbb{Z}.$$

<u>Def</u> A (pure) Hodge structure of wt k: ① ② ③ $\Leftrightarrow$ ① ②′ ③′
A polarized (pure) Hodge structure of wt k: ① ② ③ ④ $\Leftrightarrow$ ① ②′ ③′ ④

A rational/real Hodge structure needs some natural modifications on ① ④.

<u>Cor</u> For a cpt Kähler mfld X, $H^k(X;\mathbb{Z})$ is a Hodge structure of wt k.
For a cplx sm proj variety X, $H^k_{\text{prim}}(X;\mathbb{Z})$ is a polarized Hodge structure of wt k.

2. integral bilinear forms

Def. Let $L \cong \mathbb{Z}^n$ be a lattice. A bilinear form of L is a map

$$\beta : L \times L \longrightarrow (\mathbb{Z}, 1)$$

which crspds to a matrix $A_\beta = (\beta(e_i, e_j))_{ij} \in M^{n\times n}(\mathbb{Z})$, once we fix a basis of L.
The change of a basis crspds to the matrix congruence (相合).

Moreover,

β is symmetric	$\Leftrightarrow$	$A_\beta^T = A_\beta$	$M^{n\times n}_{sym}(\mathbb{Z})$
β is alternative	$\Leftrightarrow$	$A_\beta^T = -A_\beta$	$M^{n\times n}_{alt}(\mathbb{Z})$
β is non-deg	$\Leftrightarrow$	$\det A_\beta \neq 0$	
β is positive semidefinite	$\Leftrightarrow$	$\beta(v,v) \geq 0 \quad \forall v \in L$	$M^{n\times n}_{\geq 0}(\mathbb{Z})$
β is positive definite	$\Leftrightarrow$	$\beta(v,v) > 0 \quad \forall v \in L$ nonzero	$M^{n\times n}_{>0}(\mathbb{Z})$

Rmk. 1. Not all the integral quadratic form comes from the sym bilinear form.
Denote the set of int quad froms as $M^{n\times n}_{sym'}(\mathbb{Z})$, then $M^{n\times n}_{sym'}(\mathbb{Z}) \supset M^{n\times n}_{sym}(\mathbb{Z})$
e.g. the map

$$q : L \cong \mathbb{Z}^2 \longrightarrow \mathbb{Z} \qquad q(v) = v_1 v_2$$

is induced from $\beta = \begin{pmatrix} 0 & \frac{1}{2} \\ \frac{1}{2} & 0 \end{pmatrix} \in M^{2\times 2}_{sym}(\mathbb{Z}[\frac{1}{2}])$, but not $M^{2\times 2}_{sym}(\mathbb{Z})$.

$$\beta(v_1 e_1 + v_2 e_2, v_1 e_1 + v_2 e_2) = \frac{1}{2} v_1 v_2 + \frac{1}{2} v_2 \cdot v_1 = v_1 v_2$$

2. One has

$$M^{n\times n}(\mathbb{Z}) \supsetneqq M^{n\times n}_{sym}(\mathbb{Z}) \oplus M^{n\times n}_{alt}(\mathbb{Z})$$

which is fine after inverting $2 \in \mathbb{Z}$:

$$M^{n\times n}(\mathbb{Z}[\tfrac{1}{2}]) \cong M^{n\times n}_{sym}(\mathbb{Z}[\tfrac{1}{2}]) \oplus M^{n\times n}_{alt}(\mathbb{Z}[\tfrac{1}{2}])$$

One defines $GL_n(\mathbb{Z})$-action on $M^{n\times n}(\mathbb{Z})$:

$$\phi : GL_n(\mathbb{Z}) \times M^{n\times n}(\mathbb{Z}) \longrightarrow M^{n\times n}(\mathbb{Z}) \qquad \phi(P) : M \longmapsto P^T M P$$

and equiv relation $\sim_T$ on $M^{n\times n}(\mathbb{Z})$:

$$A \sim_T B \quad \text{iff} \quad \exists\, P \in GL_n(\mathbb{Z}), \quad B = P^T A P.$$

Prop. Let R be a PID. Then

$$M^{n\times n}_{alt}(R)/\sim_{T,GL_n(R)} = \left\{ \operatorname{diag}\left(\begin{pmatrix} 0 & d_1 \\ -d_1 & 0 \end{pmatrix}, \dots, \begin{pmatrix} 0 & d_k \\ -d_k & 0 \end{pmatrix}, 0, \dots, 0\right) \;\middle|\; \begin{matrix} d_i \neq 0, \\ d_1 | \cdots | d_k \end{matrix} \right\} \Big/ \sim$$

$$= \{ (d_1) \geq (d_2) \geq \cdots \geq (d_k) \gneq 0 \text{ for } 2k < n \}$$

where $(d_1, \dots, d_k) \sim (d_1', \dots, d_k') \Leftrightarrow (d_i) = (d_i') \quad \forall i$

$\Leftrightarrow \exists\, r_i \in R^\times$ s.t. $d_i' = r_i \cdot d_i$. $\forall i$

Proof. Use Gauss elimination.

E.g.

$$\begin{pmatrix} 0 & 1 & 1 \\ -1 & 0 & 1 \\ -1 & -1 & 0 \end{pmatrix} \xrightarrow{\phi\begin{pmatrix} 1 & & \\ & 1 & \\ & & -1 \\ & & & 1 \end{pmatrix}} \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix} \xrightarrow{\phi\begin{pmatrix} 1 & & \\ & 1 & \\ & & 1 \end{pmatrix}} \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$x\wedge y + x\wedge z + y\wedge z = x\wedge(y+z) + (y+z)\wedge z = (x-z)\wedge(y+z)$$

Rmk. The same method won't work for sym bilinear form, even for $M^{n\times n}_{sym}(k)$.

① divide 2 problems;

② In most cases we can't eliminate the diagonal items, even make some to be 1;

③ The congruence action is still by $GL_n(\mathbb{Z})$, not $GL_n(k)$.

Prop. Let k be a field with $2 \in k^\times$. Then

$$M^{n\times n}_{sym}(k)/\sim_{T,GL_n(k)} = \left\{ \operatorname{diag}(d_1, \dots, d_k, 0, \dots, 0) \;\middle|\; d_i \neq 0 \right\} \Big/ \sim$$

where $(d_1, \dots, d_k) \sim (d_1', \dots, d_k') \Leftarrow$ up to reordering, $\exists\, r_i \in k^\times$ s.t. $d_i' = r_i^2 \cdot d_i$. $\forall i$

E.g. In k, $2 \in k^\times$.

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \xrightarrow{\phi\begin{pmatrix} 1 & 0 \\ \frac{1}{2} & 1 \end{pmatrix}} \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \xrightarrow{\phi\begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$2xy \quad = \quad x^2 + 2x\left(-\tfrac{1}{2}x + y\right) \quad = \quad \left(\tfrac{1}{2}x + y\right)^2 - \left(-\tfrac{1}{2}x + y\right)^2$$

In $\mathbb{Z}[\frac{1}{6}]$,

$$\begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix} \xrightarrow{\phi\begin{pmatrix} 1 & -\frac{1}{3} \\ & 1 \end{pmatrix}} \begin{pmatrix} 3 & 0 \\ 0 & \frac{8}{3} \end{pmatrix}$$

$$3x^2 + 2xy + 3y^2 \quad = \quad 3\left(x - \tfrac{1}{3}y\right)^2 + \tfrac{8}{3}y^2$$